

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Distinguish qualitative and quantitative observations, choose an appropriate tool, and record a complete SI-style measurement.

ENGAGE

Two students describe the same sample as 'heavy' and '18.62 g.' Which record can another scientist compare directly, and what makes it more useful?

TEACH

- A complete quantitative record names the measured quantity and gives a number with a unit.
- Choose the tool for the quantity and needed resolution: a balance measures mass; a graduated cylinder measures liquid volume more precisely than a beaker.
- Write a space between a value and its unit symbol, use correct capitalization, and do not pluralize symbols: 6.0 mL, not 6.0mls.
- A leading zero prevents a missed decimal: write 0.45 g, not .45 g.

TERMS

qualitative: descriptive evidence without a measured number

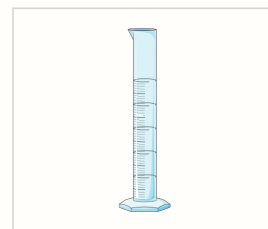
quantitative: evidence that includes a measured or counted number

resolution: the smallest increment an instrument can display or support

SI unit: a standard unit used to communicate a physical quantity



Electronic balance: mass



Graduated cylinder: liquid volume

WORKED EXAMPLE

A data table contains only '42.7.' Diagnose the record and repair it for a liquid-volume measurement.

1. Identify what is missing: the record does not name the measured quantity or unit.
2. Use the measurement context to choose the correct unit: milliliters for this liquid volume.
3. Preserve the reported precision and apply SI spacing and capitalization.

RESULT: Record the result as 'liquid volume = 42.7 mL' and identify the instrument used.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L1-1. Classify 'The solution is pale blue' and 'The solution volume is 18.5 mL' as qualitative or quantitative.

Scaffold: Look for a measured number and unit.

INDEPENDENT

L1-2. Choose the pictured tool best suited to measure 24.6 mL of water. Cite one visual feature.

INDEPENDENT

L1-3. Repair each record: (a) .45g (b) 6.0ml (c) 5 kgs.

INDEPENDENT

L1-4. A student needs the mass of a dry solid. Explain why a graduated cylinder is not the correct first tool.

EXIT TICKET

L1-5. Write one complete quantitative record for a sample with a mass display of 7.250 g.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Represent measured values in scientific notation and apply significant-figure rules to calculations.

ENGAGE

Two calculators display 39.68 and 39.680000. Does the longer display automatically make the experimental answer more precise?

TEACH

- Scientific notation has the form $a \times 10^n$, where $1 \leq a < 10$.
- Leading zeros are placeholders; zeros between nonzero digits and recorded trailing decimal zeros are significant.
- For multiplication and division, round to the fewest significant figures among measured values.
- For addition and subtraction, round to the least precise decimal place. Keep guard digits and round once at the end.

TERMS

coefficient: the number from 1 up to, but not including, 10 in scientific notation

exponent: the power of ten that records decimal movement

significant figure: a digit justified by the measurement and its recorded precision

exact number: a counted or defined value that does not limit significant figures

WORKED EXAMPLE

Calculate 12.4×3.20 and report the answer with justified precision.

1. Calculate without early rounding: $12.4 \times 3.20 = 39.68$.

2. Count significant figures: 12.4 has 3 and 3.20 has 3.

3. Round the product to 3 significant figures.

RESULT: 39.7

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L2-1. Write 0.0005070 in scientific notation and state its significant figures.

Scaffold: Move the decimal until the coefficient is between 1 and 10; preserve recorded trailing zeros.

INDEPENDENT

L2-2. State the number of significant figures in (a) 0.004060 (b) 700. (c) 1.0200 (d) 0.030.

INDEPENDENT

L2-3. Calculate $45.60 / 2.5$ and round correctly.

INDEPENDENT

L2-4. Calculate $18.42 + 3.7 + 0.006$ and round correctly.

EXIT TICKET

L2-5. A student rounds every intermediate step. State one risk and one repair.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Convert quantities by arranging conversion factors so unwanted units cancel and the target unit remains.

ENGAGE

A conversion gives the expected number, but the unwanted unit never cancels. Should the result be trusted?

TEACH

- Start with the given quantity and write its unit.
- Place each unwanted unit opposite itself so it cancels.
- Continue multiplying by equal ratios until only the target unit remains.
- Defined conversion factors, such as $1000\text{ m} = 1\text{ km}$, are exact and do not limit significant figures.

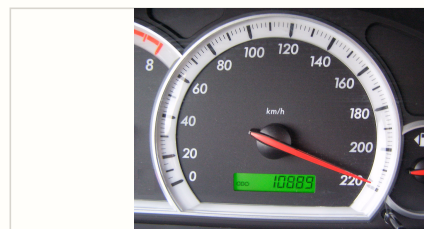
TERMS

conversion factor: a ratio of equivalent quantities that equals one

factor-label method: a method that tracks and cancels units through multiplication

target unit: the unit required in the final answer

dimensional consistency: agreement between the units in a setup and the intended quantity



A rate such as km/h combines two units; both must be tracked during conversion.

WORKED EXAMPLE

Convert 72.0 km/h to m/s.

1. Write $72.0\text{ km/h} \times (1000\text{ m} / 1\text{ km})$ so kilometers cancel.
2. Multiply by $(1\text{ h} / 3600\text{ s})$ so hours cancel.
3. Calculate $72.0 \times 1000 / 3600$ and retain three significant figures.

RESULT: 20.0 m/s

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L3-1. Convert 3.45 km to meters using a visible conversion factor.

Scaffold: Place kilometers in the denominator of the factor so they cancel.

INDEPENDENT

L3-2. Convert 0.0825 L to milliliters.

INDEPENDENT

L3-3. Convert 2.60×10^5 mg to grams and preserve significant figures.

INDEPENDENT

L3-4. Convert 18.0 m/s to km/h.

EXIT TICKET

L3-5. Explain why a setup that ends in h/s is incomplete when the target is m/s.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Calculate density, rearrange the density relationship, and use percent error to compare an experimental result with an accepted value.

ENGAGE

Two objects have different masses but the same density. What does density reveal that mass alone does not?

TEACH

- Density units combine mass and volume, commonly g/mL or g/cm³.
- Use $d = m / V$; rearrange to $m = dV$ or $V = m/d$ when needed.
- Percent error = $|\text{experimental} - \text{accepted}| / \text{accepted} \times 100\%$.
- Keep the absolute value in the numerator and use the accepted value as the denominator.

TERMS

density: mass per unit volume, $d = m / V$

accepted value: the reference value used for comparison

experimental value: the value produced by a measurement or calculation

percent error: absolute difference divided by accepted value, multiplied by 100



Read liquid volume at eye level; use the bottom of a concave meniscus.

WORKED EXAMPLE

A sample has a mass of 57.0 g and a volume of 20.0 mL. Calculate its density.

1. Select $d = m / V$ and substitute measured values with units.
2. Calculate $57.0 \text{ g} / 20.0 \text{ mL} = 2.85 \text{ g/mL}$.
3. Both measurements have three significant figures, so report three.

RESULT: 2.85 g/mL



Density and Percent Error

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L4-1. A 31.6 g sample occupies 25.5 mL. Calculate density.

Scaffold: Write $d = m / V$ before substituting values.

INDEPENDENT

L4-2. A liquid has density 1.24 g/mL and volume 25.5 mL. Calculate mass.

INDEPENDENT

L4-3. A solid has mass 14.9 g and density 2.50 g/cm³. Calculate volume.

INDEPENDENT

L4-4. An experimental value is 12.4 g/mL and the accepted value is 12.0 g/mL. Calculate percent error.

EXIT TICKET

L4-5. Explain why percent error uses the accepted value in the denominator.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Use repeated measurements and an accepted value to distinguish precision, accuracy, random variation, and systematic error.

ENGAGE

Three measurements agree closely with one another but are all higher than the accepted value. Is the method precise, accurate, both, or neither?

TEACH

- Judge precision from the spread of repeated measurements, not from one trial.
- Judge accuracy by comparing a result or mean with an accepted value.
- Random variation increases scatter; repeated trials reveal its size.
- Systematic error shifts results in a consistent direction and is not fixed by averaging more biased trials.

TERMS

precision: closeness of repeated measurements to one another

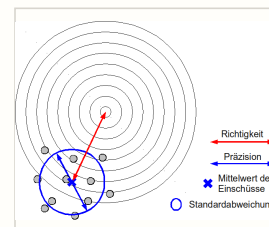
accuracy: closeness of a result to an accepted value

random error: unpredictable variation that changes from trial to trial

systematic error: a consistent offset caused by method or instrument bias



A zero offset can create a systematic mass error.



Center agreement and trial spread answer different questions.

WORKED EXAMPLE

Trials are 1.247, 1.249, and 1.248 g/mL; the accepted value is 1.200 g/mL. Evaluate the results.

1. Compare trials: the range is only 0.002 g/mL, so the measurements are precise.
2. Find the mean: $(1.247 + 1.249 + 1.248) / 3 = 1.248$ g/mL.
3. Compare with 1.200 g/mL: the mean is consistently high; percent error is 4.00%.

RESULT: Precise but not accurate relative to the accepted value; the pattern suggests a systematic high bias.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L5-1. Trials are 10.01, 10.00, and 10.02 g; accepted mass is 10.01 g. Classify accuracy and precision.

Scaffold: First compare the trials with one another; then compare their center with the accepted value.

INDEPENDENT

L5-2. Trials are 8.42, 8.43, and 8.42 mL; accepted volume is 8.00 mL. Classify accuracy and precision.

INDEPENDENT

L5-3. A student reads a meniscus from above. Name the error and give a specific repair.

INDEPENDENT

L5-4. A balance reads +0.20 g when empty. Explain why repeating trials without zeroing it does not solve the problem.

EXIT TICKET

L5-5. Explain why percent error alone cannot establish precision.



Cumulative Mixed Review

STUDENT REVIEW

Use ideas from every lesson

Name: _____ Date: _____ Period: _____

R1. Rewrite '.075kgs' as a correct SI-style record.

R2. Write 0.0008200 in scientific notation and state its significant figures.

R3. Calculate 6.20×4.1 and report the correct significant figures.

R4. Calculate $12.08 + 0.6 + 3.214$ and apply the decimal-place rule.

R5. Convert 2.75 h to seconds using dimensional analysis.



Cumulative Mixed Review

STUDENT REVIEW

Use ideas from every lesson

Name: _____ Date: _____ Period: _____

R6. A sample has mass 24.96 g and volume 10.00 mL. Calculate density.

R7. A material has density 3.60 g/mL and volume 7.25 mL. Calculate mass.

R8. Experimental density is 2.40 g/mL; accepted density is 2.50 g/mL. Calculate percent error.

R9. Trials 24.9, 25.0, and 25.1 g surround an accepted value of 25.0 g. Evaluate accuracy and precision.

R10. A conversion result has the unit kg·mL/g when the target is mL. Diagnose the setup.
